Omkar Chavan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.60
Khalsa College	Class XII (Maharashtra State Board)	2014	77.54%
Bengali Education Society's High School	Class X (Maharashtra State Board)	2012	85.82%

WORK EXPERIENCE

IIT Hyderabad | *M.Tech Research Assistant.*

Project: Layered Security for O-RAN

- Designed a Kubernetes security prototype for O-RAN APIs, using service mesh and capability tokens for **least-privilege access**.
- Scaled its authorization service to **646 requests per second**, a **5.4x throughput gain**.
- Verified deny-by-default access and outage recovery, documenting the results in a first-author manuscript.

Project: Software-Defined Perimeter Integration for O-RAN

- Studied **software-defined perimeter (SDP)** specifications and helped integrate SDP security into O-RAN, supporting protected connections between components and xApps.

IIT Madras | *Project Research Associate.*

Project: Federated Deep Learning for VNF Placement

- Built TensorFlow federated models to predict VNF resource needs without sharing provider data.
- Integrated predictions into VNF placement, reducing CPU allocation by **56%** and network delay by **55%**.
- Benchmarked federated and centralized models on synthetic and real VNF datasets.
- Co-authored the resulting paper published at **IEEE FNWF 2023**.

Tiger Analytics | *Machine Learning Engineer .*

Project: Maya Bank's HR chatbot

- Built an Azure-based HR chatbot to answer onboarding questions from internal FAQs and policies.
- Implemented retrieval-based Q&A to identify the relevant policy section for each question.
- Passed the question and retrieved context to a **T5-based model in Azure ML Studio**.
- Built Microsoft Bot Framework dialogues and integrated the chatbot with **Teams and Telegram**.

Reliance Jio Infocomm Ltd. | *Machine Learning Engineer .*

Project: Hello Jio EVA

- Built a training and inference pipeline for **intent classification** in the **Hello Jio voice/chatbot.**
- Combined attention-based LSTM embeddings with logistic regression for intent prediction.
- Developed a **spaCy entity-extraction engine** and a full-stack dashboard to analyze chatbot model performance.

Project: Video Call Assistant and Bot Platform

- Contributed to the launch of **Jio's AI-based Video Call Assistant**, unveiled at **IMC 2019**.
- Designed and implemented a reusable bot-as-a-service architecture for customer-service bots.

Reliance Jio Infocomm Ltd. | *Graduate Engineer Trainee.*

Project: Jio SAFE -- Satellite Automated Feature Extraction

- Contributed to the **end-to-end ML pipeline** for building-polygon extraction from satellite imagery using **U-Net semantic segmentation**.
- Built a training data generator that sampled **128 x 128 crops** from satellite TIFF files larger than **700 MB**.
- Improved rooftop segmentation from **58% to 71% mIoU** with the new training data pipeline.

PROJECTS

5G Control-Plane Fuzzing

- Built a NAS-aware fuzzing system with a C SCTP/NGAP proxy to test Open5GS using UERANSIM.
- Tested **5 NAS message types** and **24 mutation families**, recording 97 campaign observations.
- Found **two live cases** where registration completed despite malformed NAS identity data.

LTE Mobility and Handover Analysis with ns-3

- Simulated an LTE network with **24 UEs** and evaluated mobility at **10, 50, and 100 m/s** using ns-3.
- Benchmarked A3-RSRP, A2A4-RSRQ, and no-handover algorithms.
- Aggregate throughput decreased from **approximately 25 to 15 Mbps** as UE speed increased from 10 to 100 m/s.
- Increasing hysteresis from 1 to 6 dB reduced average handovers from **approximately 3.5 to 2.5** at 100 m/s.

Cryptanalysis of Stream and Block Ciphers

- Implemented an RC4 bias attack over 2^2 ciphertexts and **recovered a hidden six-digit passcode**.
- Brute-forced an effective **20-bit AES-128 key** and decrypted the held-out message.
- Implemented a meet-in-the-middle attack that searched **65,536 candidates per key**.

Mechanisms of Action Prediction

- Developed a multi-label classification workflow to predict biological activity from gene expression, cell

viability and dosage features.

- Performed EDA and feature engineering; compared logistic regression, SVM, XGBoost, CNN and TabNet.
- Achieved a best **log loss of 0.01627 with TabNet** across the models evaluated.

Understanding How BERT Learns to Identify Edits

- Converted AESW XML annotations into labeled sentences for academic edit detection.
- Evaluated BERT, RoBERTa and SciBERT on sentence-level edit detection.
- Compared attention with edited words using exact-match, top-3 and Jaccard measures.

Italian-to-English Translation with Attention

- Prepared **343,000 Italian-English sentence pairs** and built a **custom LSTM encoder-decoder** for Italian-to-English translation.
- Implemented dot, general and concat attention scoring functions.
- Plotted attention heatmaps to inspect which Italian words influenced each translation.

RESEARCH PAPERS & PATENT APPLICATION

VNF Placement based on Resource Usage Prediction using Federated Deep Learning Techniques

Rahul M. Verma, Krishna M. Sivalingam, **Omkar Chavan**

IEEE FNWF 2023

Layered Security for O-RAN using Service Mesh and Capability-Based Authorization

Omkar Chavan (first author), Antony Franklin

Unpublished research manuscript, 2026

Patent Title: *System and Method for API Resource Prediction* ***-- Co-inventor***, Jio Platforms Ltd.; U.S. Patent Application Publication No. **US 2023/0267012 A1** (2023).

Technical focus: Memory-efficient ML model serving using LRU caching and shared loading across worker processes.

SKILLS

Languages: Python, C, C++, JavaScript, PHP

Machine Learning & AI: TensorFlow, PyTorch, Keras, scikit-learn, NumPy, spaCy, XGBoost, TabNet, Transformers, BERT, CNNs, LSTMs, Attention Mechanisms

Backend Engineering: Flask, Django, Celery, REST APIs, SQL (MySQL/PostgreSQL), MongoDB, Redis

Cloud & DevOps: Docker, Kubernetes, AWS, Azure, Git, GitHub Actions

Telecom & Network Engineering: 5G Core, O-RAN (Open RAN), SDN-NFV, Service Mesh

RELEVANT COURSES

- **AI/ML coursework:** Representation Learning, Linear Optimization, Fraud Analytics.
- **Software and security coursework:** Advanced Data Structures and Algorithms, Network Security, Advanced Computer Networks, Topics in Wireless Networks, Cryptology.
- **Online courses:** Deep Learning Specialization (Andrew Ng, Coursera); Applied AI (Shrikant Chekuri).

POSITIONS OF RESPONSIBILITY

ISEA 5G Security Bootcamp | *RAN Security Demonstrator*

- **2026:** Guided **approximately 100 students** through a hands-on **5G air-interface security** demonstration using OpenAirInterface and Wireshark. Demonstrated how **integrity protection and AES encryption** affect the traffic visible in packet captures.
- **2025:** Volunteered on the bootcamp's **technical team**.

EXTRACURRICULAR ACTIVITIES

- **Gold Medals (2025 and 2026):** CSE department cricket tournaments.
- **Silver Medal:** Intra-college football tournament.
- **Interest:** Swimming.